ASTR 430 HW #2 Solutions

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1. Problem 6-5 from Fundamental Planetary Science

Internal vs. External heating of Pluto

<u>Assume</u>
60:40 mix of chondritic rock : ice
Albedo $A = 0.4$

Start here: Table E.16
no Pluto, but
Carbonaceous Chondrites: $4 \times 10^{-12} \frac{J}{kg\,s}$

Then $M_p = 1.3 \times 10^{22}$ kg (Table E.8)

Now ice contains no radioactivity, so

Heat generated by Pluto's interior

$$= 0.6 \cdot (1.3 * 10^{22}\ kg) \cdot (4 * 10^{-12}\ J/kg\,s)$$

$$= 3.12 * 10^{10}\ J/s \qquad (J/s = W)$$

Heat absorbed from the Sun

$$= (1-A)(\pi R_p^2) \cdot (\text{Solar Constant}) \cdot 1/a_p^2$$

where $R_p = 1.153 * 10^6\ m$ (Table E.7)

Solar Constant $= 1366\ W/m^2$ (Eq. 4.13) ← Flux at Earth

sunlight drops off as $1/a^2$

Pluto $a_p = 39.5\ AU$ (Table E.7)

so $0.6 \cdot \pi (1.153 * 10^6)^2 (1366)(1/39.5^2)$

$= 2.19 * 10^{12}\ J/s \Rightarrow$ Sunlight wins!

ratio $\dfrac{\text{radioactivity}}{\text{sunlight}} = \dfrac{3.12}{2.19} * 10^{-2} = 0.014$

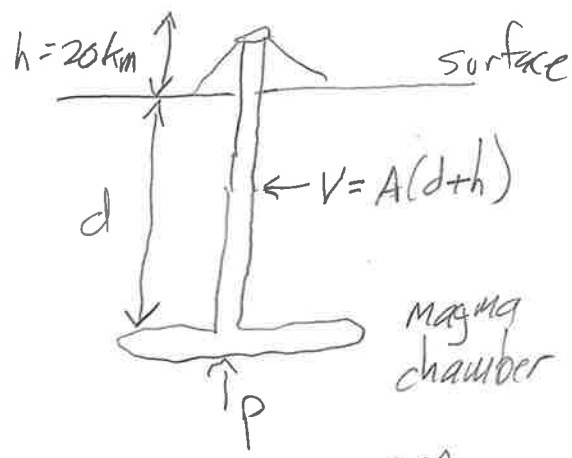
2 Problem 6-7 from FPS

Height of shield volcanoes on Mars.
= 20 km

what is depth of magma chamber if

1) pressure on chamber = ambient pressure
2) partial melt is 10% less dense than rock

Draw a picture!



Ambient Pressure means weight of the rock layers above the chamber,

So $PA = mg$ ← Force Balance

is g constant? Only if $d << R_m$

where $R_m = 3393$ km. Assume g = const and remember to check this at the end!

$$P = \frac{mg}{A} = \frac{\rho V g}{A} = \rho d g$$ ← convert to density

liquid column is supported by the same pressure,

so $P = \frac{m_e g}{A} = \frac{\rho_e V_e g}{A} = \rho_e (d+h) g$

Equate: $\rho d g = \rho_e (d+h) g$

$$\rho/\rho_e \, d = d+h$$

solve for d: $d(\rho/\rho_e - 1) = h$

$$d = \frac{h}{(\rho/\rho_e - 1)} = \frac{20\,km}{(1/0.9 - 1)} = \frac{20\,km}{0.1111}$$

$= 180\,km$ <u>below the surface</u>

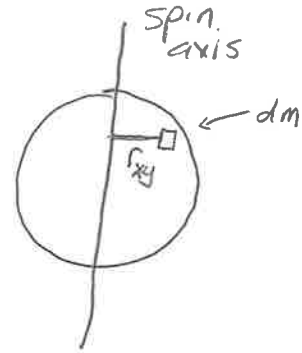
check: $d \ll R_m$ ✓ g constant is OK

like an iceberg, 90% of the liquid column is below the surface. $\frac{\rho_{ice}}{\rho_{water}} = 0.92$

HW #5

3. a)

$$dI = r_{xy}^2\, dm = (x^2+y^2)\, dm$$

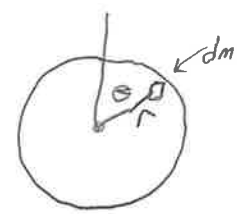


$$I = \int_M dI = \int_M (x^2+y^2)\, dm$$

The integral is easily done in spherical coordinates where $r = (x^2+y^2+z^2)^{1/2}$

$$x^2+y^2 = r^2 - z^2 = r^2 - r^2\cos^2\theta$$

$$= r^2\sin^2\theta$$



$$I = \int_M r^2\sin^2\theta\, dm$$

$$= \int_V r^2\sin^2\theta\, \rho\, dV$$

$$= \int_0^{2\pi}\int_0^{\pi}\int_0^{R} r^2\sin^2\theta\, \rho\, (r^2\sin\theta\, dr\, d\theta\, d\phi)$$

$$I = 2\pi\rho \int_0^{\pi}\int_0^{R} r^4 \sin^3\theta \, dr\, d\theta$$

$$= 2\pi\rho \left[\int_0^{\pi} \sin^3\theta \, d\theta\right]\left[\int_0^{R} r^4 \, dr\right]$$

$$= 2\pi\rho \left[\int_0^{\pi} \sin\theta\,(1-\cos^2\theta)\, d\theta\right]\left[\frac{R^5}{5}\right]$$

$$= \frac{2}{5}\pi R^5 \rho \left[-\cos\theta + \frac{\cos^3\theta}{3} \Big|_0^{\pi} \right]$$

$$= \frac{2}{5}\pi R^5 \rho \left[2 - \frac{2}{3} \right]$$

$$\boxed{I = \frac{8\pi}{15} R^5 \rho}$$

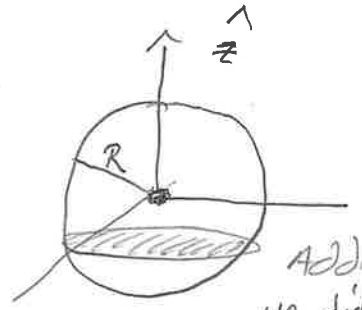
or, in terms of mass:

$$\boxed{I = \frac{2}{5} m R^2}$$

using $m = \frac{4}{3}\pi\rho R^3$

Moment of Inertia of a uniform sphere - cylindrical coordinates

$$I = \int dI = \int_M r_{xy}\, dM$$



Adding up disks to make a sphere

$$= \int_{-R}^{R}\int_{0}^{r}\int_{0}^{2\pi} r_{xy}^2\, \rho\, r_{xy}\, d\theta\, dr_{xy}\, dz$$

$$= 2\pi\rho \int_{-R}^{R} \frac{r^4}{4}\, dz \longleftarrow$$ using $r^2 = R^2 - z^2$ and expanding $(R^2 - z^2)^2$

$$= \frac{\rho\pi}{2}\left(R^4 z - \frac{2R^2 z^3}{3} + \frac{z^5}{5} \Big|_{-R}^{R} \right)$$

$$= \frac{\rho\pi}{2}\left[2R^5 - \frac{4R^5}{3} + \frac{2R^5}{5} \right]$$

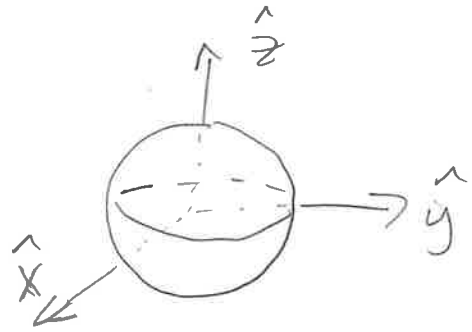
$$= \rho\pi R^5 \left[\tfrac{1}{3} R^5 + R^5/5 \right]$$

$$= \boxed{\frac{8\pi}{15}\rho R^5 = I}$$

using $M = \frac{4}{3}\pi R^3 \rho$

$$\Rightarrow \boxed{I = \frac{2}{5} M R^2}$$

Alternate solution:



$$I = \int r_{xy}^2 \, dm$$

$$I_z = \int (x^2+y^2)\,dm \rightarrow \text{around } \hat{z}$$
$$I_y = \int (x^2+z^2)\,dm \rightarrow \text{around } \hat{y}$$
$$I_x = \int (y^2+z^2)\,dm \rightarrow \text{around } \hat{x}$$

all equal by symmetry

$$3I = \int 2(x^2+y^2+z^2)\,dm = \int 2r^2\,dm$$

$$dm = 4\pi r^2 \rho \, dr$$

Sum over infinitesimal spherical shells

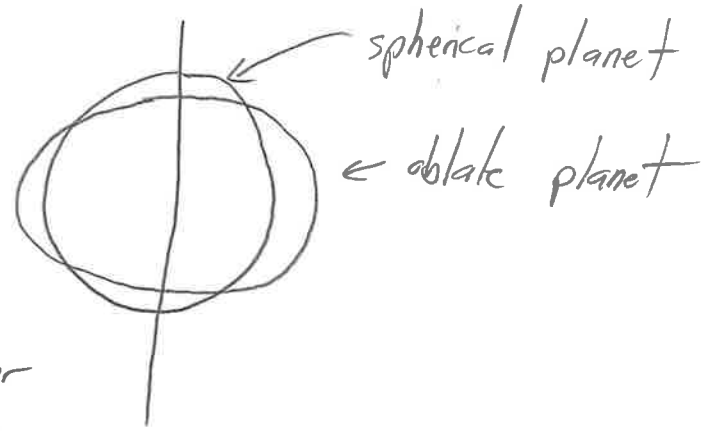
$$\Rightarrow \quad I = \frac{2}{3} 4\pi \rho \int_0^R r^4 \, dr$$

$$= \frac{8\pi}{15} \rho R^5$$

so $I = \frac{2}{5} M R^2$ as before

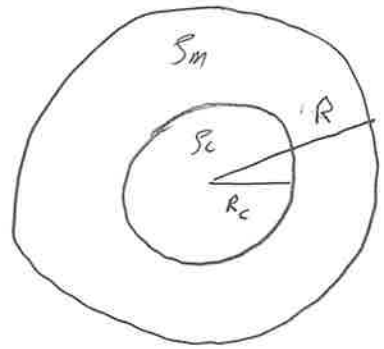
Gotta love Symmetry! ☺

b) An oblate planet has more mass located further from the spin axis than a spherical planet. Therefore, the moment of Inertia will be slightly <u>higher</u> for an oblate planet.



A planet with a heavy core will have a <u>smaller</u> moment of inertia than an equal mass uniform density planet since more mass is located near the spin axis.

c) The moment of inertia for a two component planet can be found from



$$I = I_{mantle} + I_{core}$$

$$I_{core} = \frac{8\pi}{15} R_c^5 \rho_c$$

$$I_{mantle} = \frac{8\pi}{15} R^5 \rho_m - \frac{8\pi}{15} R_c^5 \rho_m$$

(Big sphere w/ radius R minus small sphere)

$$\Rightarrow \boxed{I = \frac{8\pi}{15}\left[R_c^5(\rho_c - \rho_m) + R^5 \rho_m\right]}$$

$$\frac{8\pi}{15}\left[R_c^5(\rho_c - \rho_m) + R^5 \rho_m\right] = 0.365\, m R^2$$
$$= 0.365\, \frac{4}{3}\pi R^5 \bar{\rho}$$

$$\boxed{R_c^5(\rho_c - \rho_m) + R^5 \rho_m = 0.9125\, R^5 \bar{\rho}} \quad (1)$$

d)

$$M = \frac{4}{3}\pi \rho_c R_c^3 + \left(\frac{4}{3}\pi \rho_m R^3 - \frac{4}{3}\pi \rho_m R_c^3\right)$$

$$\boxed{M = \frac{4}{3}\pi \left[(\rho_c - \rho_m) R_c^3 + \rho_m R^3\right]}$$

use $M = \frac{4}{3}\pi R^3 \bar{\rho}$

$$\Rightarrow \boxed{R^3 \bar{\rho} = (\rho_c - \rho_m) R_c^3 + \rho_m R^3} \quad (2)$$

e)

Our two constraints are:

(1) $R_c^5 (\rho_c - \rho_m) + R^5 \rho_m = 0.9125 R^5 \bar{\rho}$

(2) $R_c^3 (\rho_c - \rho_m) + R^3 \rho_m = R^3 \bar{\rho}$

Divide (1) by R^5 and solve for R_c/R
Divide (2) by R^3 and solve for R_c/R
} Then Equate!

$$\left(\frac{R_c}{R}\right)^5 (\rho_c - \rho_m) + \rho_m = 0.9125 \bar{\rho}$$

(3) $$\frac{R_c}{R} = \left(\frac{0.9125 \bar{\rho} - \rho_m}{\rho_c - \rho_m}\right)^{1/5}$$

$$\left(\frac{R_c}{R}\right)^3 (\rho_c - \rho_m) = \bar{\rho} - \rho_m$$

$$(4) \quad \frac{R_c}{R} = \left(\frac{\bar{\rho} - \rho_m}{\rho_c - \rho_m}\right)^{1/3}$$

Equate: $\frac{R_c}{R} = \left(\frac{0.9125\bar{\rho} - \rho_m}{\rho_c - \rho_m}\right)^{1/5} = \left(\frac{\bar{\rho} - \rho_m}{\rho_c - \rho_m}\right)^{1/3}$

LHS ⇒ $\boxed{(0.9125\bar{\rho} - \rho_m)^3 (\rho_c - \rho_m)^2 = (\bar{\rho} - \rho_m)^5}$ ← RHS

This is a 5th-order polynomial in ρ_m

⇒ must solve numerically

Given $\rho_c = 7.5$ g/cm^3 (iron)

$\bar{\rho} = 3.93$ g/cm^3

	LHS	=	RHS	match?
Try $\rho_m = 3.5$:	0.0102		0.0147	not bad!
$\rho_m = 3.0$:	2.466		0.6957	Bad!
$\rho_m = 3.45$:	0.0414		0.0255	$\rho_m = 3.5$ was better
$\rho_m = 3.48$:	0.0193		0.0185	← good match!

So from (4) $\boxed{\frac{R_c}{R} = 0.482}$ Too much iron? Too little?

For the Earth, $\frac{R_c}{R} \simeq 0.53$ which is similar to that of Mars. These numbers cannot be directly compared, however, because we have ignored the fact that Mars' density will increase with depth and increasing pressure.

Mars is <u>not</u> a small version of the Earth as can be seen from comparing average densities:

$$\boxed{\begin{aligned} \bar{\rho}_{Mars} &= 3.93\ g/cm^3 \\ \bar{\rho}_{Earth} &= 5.52\ g/cm^3 \end{aligned}}$$

The favored models for Mars call for a core density <u>less</u> than the Earth's and a mantle density <u>greater</u> than Earths. This can occur if Fe is bonded to lighter elements like S (FeS) in the core, and there were more FeO in the mantle due to the fact

that Mars accreted further from the Sun than the Earth where temperatures were lower (low enough for lots of FeO to condense).

2023

Since I wrote this problem, we've landed a seismometer on Mars! We now know the core density. The core radius was found to be ≈ 1800 km, larger than expected.

$$\frac{1800 \text{ km}}{3393 \text{ km}} = 0.53 \quad \text{like Earth!}$$

This indicates that the core density is lower than expected.